LiDAR Maze 10/7

Quarter 2 Intro

Project overview

- This is the second quarter of this project
- We have some code + designs from last quarter
- More hands-on engagement this quarter
- We are building a small “car” (~10x10 inches) to autonomously solve a maze
- We must build the car and the maze
- Will be using github

Raise of hands: freshmen?



The roof-mounted unit installed on Waymo's ride-hailing vehicles contains various hardware components for autonomous driving. *Courtesy of Waymo*

Project details

- LiDAR sensor sends data to an FPGA (like a small computer, take 271)
- FPGA packages the data and sends it to arduino
- Arduino runs a real time maze-solving algorithm
- Maze solving algorithm results in motor control commands to navigate the maze
- Have to design power delivery to all components
- Design robot chassis in CAD

Integration is the difficult part, communication is key

Subteam structure

- Mechanical/Power: CAD design of chassis and power delivery system
  - Subteam Leads: Andrew, Zane
- FPGA: Write SystemVerilog code to interface with LiDAR sensor and Arduino
  - Subteam Leads: Dara, Alexa
- Software: Create maze-solving algorithms and virtual test environment
  - Subteam Leads: Akshay
- Controls: Ensure that virtual motor control commands translate to precise movements
  - Subteam Leads: Pranav

Meeting times

- Mechanical/Power and Controls team will meet **6-7PM on Mondays in ECE303**
- FPGA and software teams will meet **6-7PM on Wednesdays in ECE269**
- Show up to the meeting next week for whichever team you're interested in
- New members will have a small task to complete before they can join the project to show proficiency.
- Any questions at this point?

Attendance

- More strict attendance policy
- Miss one meeting: warning
- Miss two meetings: talk to lead
- Miss three meetings: kicked from project
- Please communicate your absences with your subteam lead
- (obvious exemptions if you’re sick/injured, etc.)

Other expectations

- Take initiative, be hands-on
- Dont worry about breaking something or feeling like you’re overstepping
- Ask questions to your lead, that’s why we are here
- Communicate updates
- Please do not be scared to talk in the discord or dm people!



Any questions?

- Leads will stay for a few minutes